Abstract

Accurate real estate valuation is crucial for financial institutions, investors, and policymakers. Traditional methods, such as hedonic pricing models and basic machine learning approaches, often fail to capture the complex spatial dependencies inherent in property data. This paper proposes a Graph Convolutional Network (GCN)-based model to enhance real estate price prediction by leveraging spatial relationships between properties. The model preprocesses data using label encoding and standard scaling while constructing an adjacency matrix to represent property interconnectivity. By incorporating graph convolutional layers, the model effectively captures spatial dependencies, resulting in improved accuracy over traditional methods. Experimental results demonstrate that the proposed GCN model outperforms conventional machine learning models in predictive performance, showcasing its potential for robust and scalable real estate valuation. The proposed approach contributes to the advancement of AI-driven real estate assessment, ensuring more reliable and data-driven decision-making in the field.